



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

**SENIOR CERTIFICATE EXAMINATIONS/
NATIONAL SENIOR CERTIFICATE EXAMINATIONS
SENIORSERTIFIKAAT-EKSAMEN/
NASIONALE SENIORSERTIFIKAAT-EKSAMEN**

TECHNICAL MATHEMATICS P1/TEGNIJSE WISKUNDE V1

2023

FINAL MARKING GUIDELINES/FINALE NASIENRIGLYNE

MARKS/PUNTE: 150

Marking Codes/Nasienkodes	
A	Accuracy/Akkuraatheid
AD	Additional Notes on ADDENDUM/ Bykomende notas op ADDENDUM
CA	Consistent Accuracy/Volgehoue Akkuraatheid
M	Method/Metode
R	Rounding/Afronding
NPR	No Penalty for Rounding/Geen Penaliserings vir Afronding nie
NPU	No Penalty for Units omitted/Geen Penaliseering vir Eenhede Weggelaat nie
S	Simplification/Vereenvoudiging
SF	Substitution in Correct Formula/Vervanging in Korrekte Formule

**These marking guidelines consist of 18 pages.
Hierdie nasienriglyne bestaan uit 18 bladsye.**

NOTE:

- If a candidate answers a question TWICE, only mark the FIRST attempt.
- Consistent accuracy applies in all aspects of the marking guidelines where indicated.
- # Shows questions where Tolerance Range will be applied:
Q 1.2.2 ; Q 3.3.2 ; Q 5.3 ; Q 9.2

LET WEL:

- Indien 'n kandidaat 'n vraag TWEE keer beantwoord, sien slegs die EERSTE poging na.
- Volgehoue akkuraatheid is deurgaans op alle aspekte van die nasienriglyne waar aangedui.
- # Toon vare waar Toleransie Wydte (Verdraagsaamheids omvang) toegepas word:
V 1.2.2 ; V 3.3.2 ; V 5.3 ; V 9.2

QUESTION/VRAAG 1

1.1.1	$\frac{1}{2}x(2x-1)=0$ $x=0$ or/of $x=\frac{1}{2}$	$\checkmark x=0$ A $\checkmark x=\frac{1}{2}$ A (2)
1.1.2	$-x(6-x)=4$ $x^2-6x-4=0$ $x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$ $=\frac{-(-6)\pm\sqrt{(-6)^2-4(1)(-4)}}{2(1)}$ $x\approx-0,61$ or/of $x\approx 6,61$	AD $\checkmark$ standard form / A standaardvorm $\checkmark$ SF CA $\checkmark x\approx-0,61$ CA $\checkmark x\approx 6,61$ CA R (4)
1.1.3	$(2-x)(x+5)>0$ OR/OF $(x-2)(x+5)<0$ $-5<x<2$ OR/OF $x\in(-5;2)$ OR/OF $x>-5$ and/en $x<2$	$\checkmark$ critical values/ A kritiese waarde $\checkmark$ correct notation/ A korrekte notasie (2)
1.2.1	$y+x-10=0$ $y=-x+10$	$\checkmark y=-x+10$ A (1)
1.2.2 #	$y=-x+10$ $x^2-xy+y^2=28$ $x^2-x(-x+10)+(-x+10)^2=28$ $x^2+x^2-10x+x^2-20x+100=28$ $\therefore 3x^2-30x+72=0$ or/of $x^2-10x+24=0$ $(x-6)(x-4)=0$ OR/OF $x=\frac{-(-10)\pm\sqrt{(-10)^2-4(1)(24)}}{2(1)}$ $\therefore x=6$ or/of $x=4$ $\therefore y=-6+10=4$ or/of $y=-4+10=6$	$\checkmark$ subst./ verv. CA $\checkmark$ expanding/ uitbrei CA $\checkmark$ standard form / CA standaardvorm $\checkmark$ factors/formula CA /faktore/formule $\checkmark$ both x-values/albei CA x-waardes $\checkmark$ both y-values/albei

OR/OF	y-waardes OR/OF	CA
$x = -y + 10$		
$x^2 - xy + y^2 = 28$		
$(-y + 10)^2 - y(-y + 10) + y^2 = 28$		
$y^2 - 20y + 100 + y^2 - 10y + y^2 = 28$	✓ subst/verv	
$3y^2 - 30y + 72 = 0$ or / of $y^2 - 10y + 24 = 0$	✓ expanding	CA
	✓ standard form / standaardvorm	CA
$(y - 4)(y - 6) = 0$ OR / OF $y = \frac{-(-10) \pm \sqrt{(-10)^2 - 4(1)(24)}}{2(1)}$	✓ factors/formula /faktore/formule	CA
$\therefore y = 4$ or/of $y = 6$	✓ both y-values/ beide y--waardes	CA
$\therefore x = -4 + 10 = 6$ or/of $x = -6 + 10 = 4$	✓ both x-values/ beide x-waardes	CA
		(6)

1.3.1	$P = \frac{F}{A}$ $PA = F$ $A = \frac{F}{P}$	✓ subject/ onderwerp A (1)
1.3.2	$A = \frac{F}{P}$ $A = \frac{25 \times 10^3}{25\,984\,480,5} \text{ m}^2$ $\approx 0,000962 \text{ m}^2$ $\approx 9,62 \times 10^{-4} \text{ m}^2$ OR/OF $P = \frac{F}{A}$ $25\,984\,480,5 = \frac{25 \times 10^3}{A}$ $A \approx 0,000962 \text{ m}^2$ $A \approx 9,62 \times 10^{-4} \text{ m}^2$	AD ✓ SF CA ✓ S CA ✓ Scientific notation/ Wetenskaplike notasie CA OR/OF ✓ SF A ✓ S CA ✓ Scientific notation/ Wetenskaplike notasie CA NPU NPR (3)

1.4.1	$=1000111_2 - 10011_2$ $= 110100_2$ OR/OF 1000111₂ and/en 10011₂ <table><tr><td>2⁶</td><td>2⁵</td><td>2⁴</td><td>2³</td><td>2²</td><td>2¹</td><td>2⁰</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td></td><td></td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td></tr></table> $64 + 4 + 2 + 1 = 71$ $16 + 2 + 1 = 19$ $\therefore 71 - 19 = 52$ $52 = 32 + 16 + 4 = 110100_2$ AD ✓ M Substitution/ vervanging ✓ 110100₂ OR/OF ✓M ✓ 110100₂ AO: Full Marks/volpunte A CA A CA (2)	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰	1	0	0	0	1	1	1			1	0	0	1	1
2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰																
1	0	0	0	1	1	1																
		1	0	0	1	1																
1.4.2	$110100_2 = 2^5 + 2^4 + 2^2$ $= 52$	✓ decimal form/ desimale vorm	CA (1)																			
			[22]																			

QUESTION/VRAAG 2

2.1.1	$1 - 3k = 0$ $k = \frac{1}{3}$	✓ value of / waarde van k A (1)
2.1.2	$k = 4$	✓ value of / waarde van k A (1)
2.2.1	< 0 OR / OF is negative / negatief OR / OF less than zero/minder as nul	✓ < 0 or/of is neg or/of less than zero/minder as nul A (1)
2.2.2	$4x^2 + 3x + p = 0$ $\Delta = b^2 - 4ac$ $= (3)^2 - 4(4)(p)$ $= 9 - 16p$ $9 - 16p < 0$ $p > \frac{9}{16}$	AD ✓ SF A ✓ S CA ✓ $\Delta < 0$ A ✓ value of /waarde van p CA (4)
		[7]

QUESTION/VRAAG 3

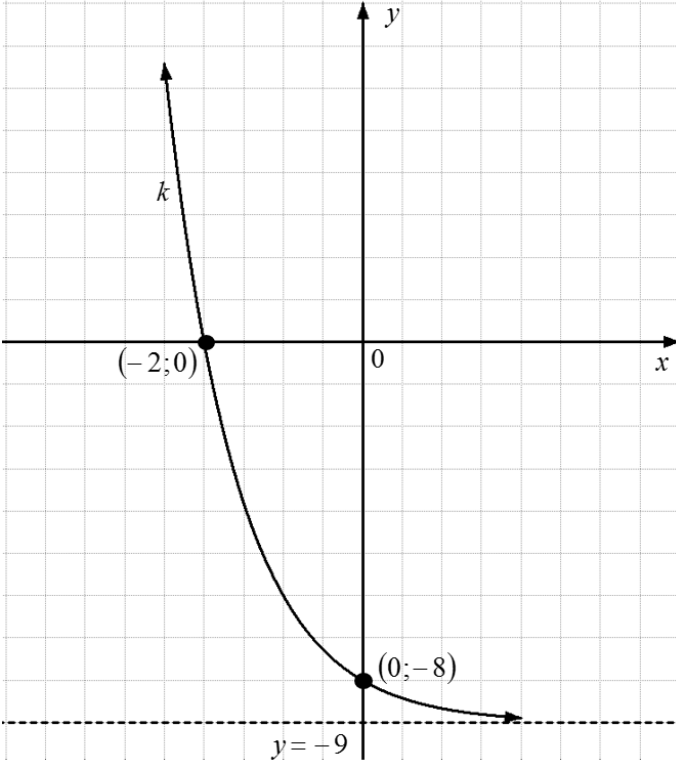
3.1.1	$7(3x)^0 = 7(1) = 7$	✓ 7(1) or /of 7	A (1)
3.1.2	$\sqrt{8}(\sqrt{242} - \sqrt{72})$ $= \sqrt{2 \times 4}(\sqrt{2 \times 121} - \sqrt{2 \times 36})$ $= 2\sqrt{2}(11\sqrt{2} - 6\sqrt{2})$ OR/OF $= 2\sqrt{2}(5\sqrt{2})$ OR/OF $= 2\sqrt{2}(11\sqrt{2}) - 2\sqrt{2}(6\sqrt{2})$ OR/OF $= 22 \times 2 - 12 \times 2$ $= 20$	✓ $\sqrt{2 \times 121}$ or/of $11\sqrt{2}$ A ✓ $\sqrt{2 \times 36}$ or/of $6\sqrt{2}$ A ✓ S CA AO: Zero Marks/ Nul punte (3)	
3.1.3	$\frac{9^{n-1} \times 27^{3-2n}}{81^{2-n}}$ $= \frac{3^{2(n-1)} \times 3^{3(3-2n)}}{3^{4(2-n)}}$ $= \frac{3^{2n-2} \times 3^{9-6n}}{3^{8-4n}}$ $= 3^{2n-2+9-6n-8+4n}$ $= \frac{1}{3}$ OR/OF 3^{-1}	✓ prime base/ <i>priem-basis</i> A ✓ expanding/ <i>uitbrei</i> CA ✓ exponent properties CA <i>/eksponent</i> <i>eienskappe</i> ✓ $\frac{1}{3}$ OR/OF 3^{-1} CA (4)	

3.2	$\log(x+2) - \log x = 1$ $\log \frac{x+2}{x} = 1$ OR/OF $\log \frac{x+2}{x} = \log 10$ $\frac{x+2}{x} = 10^1$ $10x = x + 2$ $\therefore x = \frac{2}{9}$ OR/OF $\log(x+2) - \log x = 1$ $\log(x+2) = \log 10 + \log x$ $\log(x+2) = \log 10x$ $x+2 = 10x$ $\therefore x = \frac{2}{9}$	✓ log properties/ <i>eienskappe</i> ✓ exponent form/ <i>eksponent vorm</i> ✓ S ✓ $x = \frac{2}{9}$ OR/OF ✓ $\log 10 + \log x$ ✓ log property/ <i>eienskappe</i> ✓ S ✓ $x = \frac{2}{9}$ CA (4)
3.3.1	z lies in the 4 th Quadrant / lê in 4de kwadrant	AD ✓ correct quadrant/ <i>korrekte kwadrant</i> A (1)
3.3.2 #	$z = 5 - 5i$ $r = \sqrt{5^2 + (-5)^2}$ $= 5\sqrt{2}$ $\tan \theta = -\frac{5}{5} = -1$ ref / verw. $\angle = 45^\circ$ $\theta = 315^\circ$ $z = 5\sqrt{2} \text{ cis } 315^\circ$ OR/OF $2\sqrt{5} \angle 315^\circ$ OR/OF $z = 5\sqrt{2} (\cos 315^\circ + i \sin 315^\circ)$	AD ✓ modulus(r) ✓ reference $\angle$ / verw $\angle$ ✓ argument ✓ polar form / <i>polêre vorm</i> [Accept radians/ Aanvaar radiale] AO: Full Marks/volpunte (4)
3.4	$m = 3i(2i - 5) + 7 - ni$ $m + ni = 6i^2 - 15i + 7$ $m + ni = 6(-1) - 15i + 7$ $m = 1$ $n = -15$	✓ expanding ✓ $i^2 = -1$ ✓ value of / waarde van m ✓ value of /waarde van n CA CA (4)
		[21]

QUESTION/VRAAG 4

4.1.1	A: $x = -2$ B: $x = 4$	AD ✓ x -coordinate of / x -koordinaat van A A ✓ x -coordinates of / x -koordinaat van B A (2)
4.1.2	$x = \frac{-2 + 4}{2}$ $= 1$ $y = -(1)^2 + 2(1) + 8 \text{ OR/OF } y = -(1-4)(1+2)$ $= 9$ $D(1; 9)$ OR/OF $y = -(x^2 - 2x - 8)$ $y = -x^2 + 2x + 8$ $x = -\frac{b}{2a}$ $= -\frac{2}{2(-1)}$ $= 1$ $y = -(1)^2 + 2(1) + 8 \text{ OR/OF } y = -(1-4)(1+2)$ $= 9$ $D(1; 9)$ OR/OF $f'(x) = 0$ $-2x + 2 = 0$ $x = 1$ $y = -(1)^2 + 2(1) + 8 \text{ OR/OF } y = -(1-4)(1+2)$ $= 9$ $D(1; 9)$	✓ M Midpoint/ middelpunt A ✓ $x = 1$ CA ✓ $y = 9$ CA OR/OF ✓ M formula/ formule A ✓ $x = 1$ CA ✓ $y = 9$ CA OR/OF ✓ M derivative/ afgeleide A ✓ $x = 1$ CA ✓ $y = 9$ CA (3)
4.1.3	(a) $y \leq 9$ OR/OF $y \in (-\infty; 9]$	✓ answer/antwoord from/vanaf Q/V 4.1.2 CA (1)
	(b) $x = 0$ and / en	AD ✓ $x = 0$ A ✓ $y = 9$ CA

		from/vanaf Q/V 4.1.2 (2)
4.1.4	$g(x) = \frac{k}{x} + 9$ $0 = \frac{k}{-2} + 9$ $k = 18$	✓ subst.asymptote/verv. <i>asimptoot</i> CA ✓ subst/verv A CA ✓ value of/waarde van <i>k</i> CA (3)
4.1.5	$-2 \leq x < 0$ OR/OF $x \in [-2 ; 0)$	✓ endpoints/eindpunte A ✓ correct notation/notasie A (2)
4.2.1	$h(x) = 2x + c$ $0 = 2(-3) + c$ $6 = c$	✓ subst/verv S A ✓ value of/waarde van <i>c</i> CA (2)
4.2.2	(a) $M(-6; 0)$	✓ coordinates/coordinate CA from/ vanaf Q/V 4.2.1 (1)
	(b) $p(x) = \sqrt{36 - x^2}$	36 or/of 6^2 CA from/ vanaf Q/V 4.2.1 or 4.2.2 (a) (1)
	(c) $-6 \leq x \leq 6$ OR/OF $x \in [-6 ; 6]$	✓ endpoints/eindpunte CA ✓ correct notation/notasie A (2)
4.3.1	$y = -9$	✓ asymptote / <i>asimptoot</i> A (1)
4.3.2	$k(x) = \left(\frac{1}{3}\right)^x - 9$ $0 = \left(\frac{1}{3}\right)^x - 9$ $3^{-x} = 3^2$ $x = -2$ $k(x) = \left(\frac{1}{3}\right)^0 - 9$ $y = -8$	 ✓ $3^{-x} = 3^2$ A ✓ $x = -2$ CA ✓ $y = -8$ A (3)

4.3.3		AD ✓ asymptote/asimptote CA ✓ both intercepts/ beide afsnitte CA ✓ exponent shape/ eksponent vorm CA (3)
		[26]

QUESTION/VRAAG 5

5.1	$A = P(1+i)^n$ $= 75(1+26\%)^3$ $\approx 150,03^\circ\text{C}$	AD ✓ F A ✓ SF A ✓ temp CA NPU NPR (3)
5.2.1	R 27 000	✓ 27 000 A (1)
5.2.2	$A = P(1-i)^n$ $27\,000 = 81\,000(1-0,2)^n$ $\frac{27\,000}{81\,000} = (0,8)^n$ $n = \log_{0,8} \frac{27\,000}{81\,000}$ OR / OF $n = \log_{(1-0,2)} \frac{1}{3}$ $n \approx 4,92 \text{ years/jare}$	✓ SF CA ✓ S CA ✓ log form / vorm CA ✓ 4,92 CA (4)
5.3 #	<u>Amount on / bedrag op 1 Feb 2025:</u> $A_1 = R30\,000 \left(1 + \frac{10\%}{12}\right)^{12 \times 2} + R20\,000$ $\approx R56\,611,72884$ <u>Amount on / bedrag op 1 Feb 2026:</u> $A_2 = [56\,611,72884] \times \left(1 + \frac{12\%}{2}\right)^2 + R10\,000$ $\approx R73\,608,93853$ <u>Amount on / bedrag op 1 Feb 2028:</u> $A_3 = 73\,608,93853 \times \left(1 + \frac{12\%}{2}\right)^{2 \times 2}$ $\approx R92\,929,59$ The municipality will NOT will be able to save enough money for the upgrade / Die munisipaliteit sal NIE genoeg geld gespaar het vir die opgradering nie. OR/OF	✓ $R30\,000 \left(1 + \frac{10\%}{12}\right)^{12 \times 2}$ A A ✓ + R20 000 ✓ $A_1 \times \left(1 + \frac{12\%}{2}\right)^2$ CA A ✓ + R10 000 ✓ $A_2 \times \left(1 + \frac{12\%}{2}\right)^{2 \times 2}$ CA CA ✓ conclusion/ gevolgtrekking OR/OF

	$A = R30\,000 \left(1 + \frac{10\%}{12}\right)^{2 \times 12} \left(1 + \frac{12\%}{2}\right)^{3 \times 2}$ $+ R20\,000 \left(1 + \frac{12\%}{2}\right)^{2 \times 3}$ $+ R10\,000 \left(1 + \frac{12\%}{2}\right)^{2 \times 2}$ $A \approx R92\,929,59$ The municipality will NOT be able to save enough money for the upgrade / <i>Die munisipaliteit sal NIE genoeg geld gespaar het vir die opgradering nie.</i> OR/OF $A = \left[\left\{ R30\,000 \left(1 + \frac{10\%}{12}\right)^{12 \times 2} + R20\,000 \right\} \times \left(1 + \frac{12\%}{2}\right)^2 + R10\,000 \right]$ $\times \left(1 + \frac{12\%}{2}\right)^{2 \times 2}$ $A \approx R92\,929,59$ The municipality will NOT be able to save enough money for the upgrade / <i>Die munisipaliteit sal NIE genoeg geld gespaar het vir die opgradering nie.</i>	✓ $R30\,000 \left(1 + \frac{10\%}{12}\right)^{12 \times 2}$ A A ✓ + R20 000 A ✓ × $\left(1 + \frac{12\%}{2}\right)^2$ A ✓ + R10 000 ✓ × $\left(1 + \frac{12\%}{2}\right)^{2 \times 2}$ CA ✓ conclusion/ gevolgtrekking CA OR/OF ✓ $R30\,000 \left(1 + \frac{10\%}{12}\right)^{12 \times 2}$ A A ✓ + R20 000 A ✓ × $\left(1 + \frac{12\%}{2}\right)^2$ A ✓ + R10 000 CA ✓ × $\left(1 + \frac{12\%}{2}\right)^{2 \times 2}$ CA ✓ conclusion/ gevolgtrekking (6) NPR
		[14]

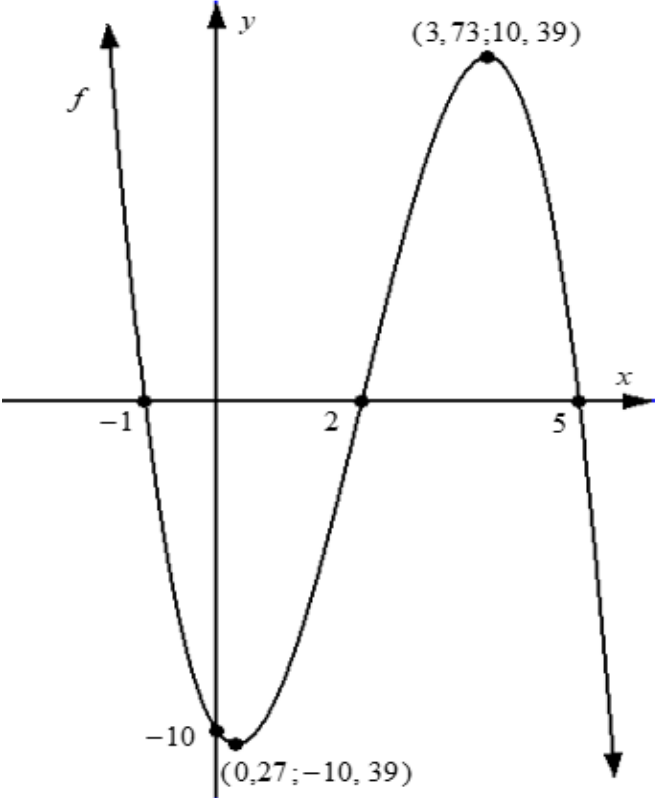
QUESTION/VRAAG 6

6.1	$f(x) = \frac{7}{2}x + 5$ $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ $= \lim_{h \rightarrow 0} \frac{\frac{7}{2}(x+h) + 5 - \left(\frac{7}{2}x + 5\right)}{h}$ $= \lim_{h \rightarrow 0} \frac{\frac{7}{2}x + \frac{7}{2}h + 5 - \frac{7}{2}x - 5}{h}$ $= \lim_{h \rightarrow 0} \frac{\frac{7}{2}h}{h}$ $= \lim_{h \rightarrow 0} \frac{7}{2}$ $\therefore f'(x) = \frac{7}{2} \text{ OR/OF } 3,5$	AD ✓ definition/definisie A ✓ SF CA Penalty of one mark for incorrect notation <i>Penaliseer een punt indien notasie foutief is.</i> ✓ S CA AO one mark/slegs een punt ✓ $\frac{7}{2}$ OR/OF 3,5 CA (5)
6.2.1	$f(x) = -8\pi$ $f'(x) = 0$	AD ✓ 0 A (1)
6.2.2	$y = \frac{x^4 + 9x}{x^2}$ $= x^2 + 9x^{-1}$ $\frac{dy}{dx} = 2x - 9x^{-2} \text{ OR / OF } \frac{dy}{dx} = 2x - \frac{9}{x^2}$	AD ✓ S A ✓ $2x$ CA ✓ $-9x^{-2}$ or / of $-\frac{9}{x^2}$ CA (3)
6.2.3	$D_x \left[(\sqrt{x} + 8x)^2 \right]$ $= D_x \left[\left(x^{\frac{1}{2}} + 8x \right)^2 \right]$ $= D_x \left[x + 16x^{\frac{3}{2}} + 64x^2 \right]$ $= 1 + 24x^{\frac{1}{2}} + 128x \text{ OR / OF } = 1 + 24\sqrt{x} + 128x$	AD ✓ $x^{\frac{1}{2}}$ A ✓ expanding/uitbrei CA ✓ 1 CA ✓ $24x^{\frac{1}{2}}$ OR/OF $24\sqrt{x}$ CA ✓ $128x$ CA (5)

6.3.1	$g(x) = 3x^2 + 9x$ $g'(x) = 6x + 9$	✓ derivative of /afgeleide v A (1)
6.3.2	$g'(x) = 6x + 9$ $\therefore m_T = 6(-3) + 9$ $= -9$	✓ subst.into derivative / vervang in afgeleide ✓ -9 CA CA (2)
6.3.3	$y = 3(-3)^2 + 9(-3)$ $\therefore y = 0$ $y = mx + c$ OR/OF $y - y_1 = m(x - x_1)$ $0 = -9(-3) + c$ $y - 0 = -9(x + 3)$ $c = -27$ $y - 0 = -9x - 27$ $\therefore y = -9x - 27$	✓ y- coordinate / koördinaat A ✓ SF CA ✓ eqn. of tangent / vergelyking v raaklyn CA (3)
		[20]

QUESTION/VRAAG 7

7.1	$(0; -10)$	AD ✓ 0 ✓ -10	A A (2)
7.2	$f(-1) = -(-1)^3 + 6(-1)^2 - 3(-1) - 10$ $= 0$ $\therefore (x + 1)$ is a factor of/is a <i>factore van</i> f	✓ substitution/ <i>vervanging</i> ✓ 0	A A (2)
7.3	x -intercepts; $y = 0$ $-(x - 2)(x^2 - 4x - 5) = 0$ $-(x - 2)(x + 1)(x - 5) = 0$ $\therefore x = 2$ or/of $x = -1$ or/of $x = 5$	AD ✓ $x = 2$ ✓ $x = -1$ and/en $x = 5$ AO: full marks/volpunte	A A (2)
7.4	$f'(x) = -3x^2 + 12x - 3$ $f'(x) = -3x^2 + 12x - 3 = 0$ $= x^2 - 4x + 1 = 0$ $x = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(1)}}{2(1)}$ $\therefore x \approx 0,27$ or/of $x \approx 3,73$ $y = f(0,27) = -(0,27)^3 + 6(0,27)^2 - 3(0,27) - 10$ $\approx -10,39$ $y = f(3,73) = -(3,73)^3 + 6(3,73)^2 - 3(3,73) - 10$ $\approx 10,39$ $\therefore (0,27; -10,39)$ and/en $(3,73; 10,39)$	✓ derivative/ <i>afgeleide</i> ✓ equating derivative to 0/ <i>stel afgeleide gelyk aan 0</i> ✓ SF ✓ both values of / <i>beide waardes van</i> x ✓ both values of / <i>beide waardes van</i> y	A A CA CA CA (5)

7.5		AD ✓ y-intercept/ afsnit CA ✓ all x-intercepts/ al die x- afsnitte CA ✓ both turning points/ beide draaipunte CA ✓ cubic shape/ kubiek vorm CA
7.6	$0,27 < x < 3,73$ OR/OF $x \in (0,27 ; 3,73)$ OR/OF $x > 0,27$ and/en $x < 3,73$	✓ critical values / kritiesewaardes CA ✓ correct notation / korrekte notasie CA
		(4)
		(2)
		[17]

QUESTION/VRAAG 8

8.1	$r^2 = 9 - h^2$ $r = \sqrt{9 - h^2}$	✓ Applying / <i>toepassing</i> Pythagoras Theorem/ <i>Stelling</i> A ✓ $r = \sqrt{9 - h^2}$ A (2)
8.2	$V = \frac{1}{3} \pi r^2 h$ $V(h) = \frac{1}{3} \pi (\sqrt{9 - h^2})^2 h$ $= \frac{1}{3} \pi (9 - h^2) h$ $= 3\pi h - \frac{1}{3} \pi h^3$	✓ SF CA ✓ S CA (2)
8.3	$V(h) = 3\pi h - \frac{1}{3} \pi h^3$ $V'(h) = 3\pi - \pi h^2$ For maximum/vir maksimum: $V'(h) = 0$ $\therefore 3\pi - \pi h^2 = 0$ $-\pi h^2 = -3\pi$ $h^2 = 3$ $h = \sqrt{3} \text{ cm}$ OR/OF $h \approx 1,73 \text{ cm}$	✓ derivative/ <i>afgeleide</i> A ✓ equating derivative to 0 / <i>stel afgeleide aan 0</i> A ✓ S CA ✓ value of/waarde van h CA NPU NPR (4)
		[8]

QUESTION/VRAAG 9

9.1.1	$\int \frac{3}{x} dx$ $= 3 \ln x + C$	$\checkmark 3 \ln x$ $\checkmark C$	A A (2)
9.1.2	$\int \left(-\frac{3x}{x^5} + \sqrt[5]{x^3} \right) dx$ $= \int \left(-3x^{-4} + x^{\frac{3}{5}} \right) dx$ $= \frac{-3x^{-3}}{-3} + \frac{x^{\frac{8}{5}}}{\frac{8}{5}} + C$ $= x^{-3} + \frac{5}{8}x^{\frac{8}{5}} + C \quad \text{OR/OF} \quad = \frac{1}{x^3} + \frac{5}{8}x^{\frac{8}{5}} + C$	AD $\checkmark -3x^{-4}$ $\checkmark x^{\frac{3}{5}}$ $\checkmark x^{-3} \quad \text{OR/OF} \quad \frac{1}{x^3}$ $\checkmark \frac{5}{8}x^{\frac{8}{5}} \quad \text{OR/OF} \quad \frac{x^{\frac{8}{5}}}{\frac{8}{5}}$	A A CA CA (4)

9.2 #	$A_1 = - \int_0^3 f(x) dx$ $= - \int_0^3 (x^2 - 5x) dx$ $= - \left[\frac{x^3}{3} - \frac{5}{2} x^2 \right]_0^3$ $= - \left[\left[\frac{(3)^3}{3} - \frac{5}{2} (3)^2 \right] - \left[\frac{(0)^3}{3} - \frac{5}{2} (0)^2 \right] \right]$ $= 13,5 \text{ OR/OF } \frac{27}{2} \text{ units}^2 / \text{eenhede}^2$ $A_2 = \int_5^7 (x^2 - 5x) dx$ $= \left[\frac{x^3}{3} - \frac{5}{2} x^2 \right]_5^7$ $= \left[\frac{(7)^3}{3} - \frac{5}{2} (7)^2 \right] - \left[\frac{(5)^3}{3} - \frac{5}{2} (5)^2 \right]$ $= \frac{38}{3} \text{ OR/OF } \approx 12,67 \text{ units}^2 / \text{eenhede}^2$ $\therefore A_1 - A_2 = 13,5 - 12,67 \text{ units}^2 / \text{eenhede}^2$ $\approx 0,83 \text{ units}^2 / \text{eenhede}^2$ $\text{OR / OF } = \frac{5}{6} \text{ units}^2 / \text{eenhede}^2$	AD ✓ M Area notation using integrals/ Oppervlakte- notasie met gebruik van integrale A ✓ $\frac{x^3}{3} - \frac{5}{2} x^2$ A ✓ Substitution/ Vervanging CA ✓ 13,5 OR/OF $\frac{27}{2}$ CA ✓ M Area notation using integrals/ Oppervlakte- notasie met gebruik van integrale A ✓ Substitution/ vervanging CA ✓ $\frac{38}{3}$ OR/OF 12,67 CA ✓ Subtracting Areas/ Areas aftrek M A ✓ 0,83 OR/OF $\frac{5}{6}$ CA NPR AO: 3 marks/punte (9) [15]

TOTAL/TOTAAL: 150